

Master Troubleshooting Guide

Rate the questions below 1-4.

- 1) I have seen it
- 2) I understand it
- 3) I perform it
- 4) I can teach it.

Also below each item describe the steps in the correct order to trouble shoot each item and describe all the components of each system.

1) Incandescent 1 2 3 4

Troubleshooting: check continuity or replace bulb with a known good bulb, Check power, check light switch, check socket/socket eye, check for thermal protector
Components: Fixture, lamp, socket, wire, switch, thermal protector

2) Fluorescent/CF 1 2 3 4

Troubleshooting: check lamps ends to see if they are black, if lamps are black change them, if not black check for power with meter, if there is power then change the ballast or igniter if all lamps are out. If some lamps are working then check the wiring/sockets. If there is no power check for a switch, then trace it back to the first junction box. On many T5 style fixtures if the lamps have been removed and replaced, the power to the ballast must be reset in order to fire the lamps.
Components: Fixture, lamp, ballast, igniter, sockets, wire, switch.

3) HID 1 2 3 4

Troubleshooting: test socket voltage with meter (refer to socket voltage chart). If there is proper voltage then change the lamp. If there is not proper voltage at the socket test for proper line voltage with meter. If there is not proper line voltage then test the fuses and check for fuses at the base pole. If the line voltage is correct then test the capacitor voltage from the ballast. If the correct voltage is coming from the ballast to the capacitor but the correct voltage is not leaving the capacitor then change the capacitor. If the ballast is not sending the correct voltage then change the ballast kit. If the capacitor or ballast is not the problem then check the socket. If the socket is not the problem then change the igniter. If you replace a ballast kit you should always put a fresh lamp in. Components: Fixture, lamp, ballast kit (core and coil, capacitor, igniter), socket, wire, fuses, photocell

4) Neon 1 2 3 4

Troubleshooting: EXPOSED NEON: Listen/look/feel for voltage in the glass, if you do

not hear/see/feel voltage on any of the units go back to the transformer and check switches/incoming power. If there is good power check for broken neon (first piece of neon off the transformer). Change the transformer if it will not go into bypass mode/send power when the secondary leads are disconnected. If one or more units are not vibrating as much as other neon units, bypass those units and verify function of other units. If bypassing does not work then check for GTO burns. Replace GTO as needed.

CHANNEL NEON: Listen/look/feel for voltage in the glass thru the drain holes, if you do not hear/see/feel voltage on any of the units go back to the transformer and check switches/incoming power. If there is good power check from broken neon (first piece of neon off the transformer). Change the transformer if it will not go into bypass mode/send power when the secondary leads are disconnected. To troubleshoot the neon go to the center unit and go from GTO to ground on each of the leads off the unit. ½ the sign should light up and the direction that did not light up should then be split in half and repeat the narrowing process until the bad unit/units are found. If bypassing does not work then check for GTO burns. Replace GTO as needed.

Components: Glass units, booties, GTO, transformer, PK housings, glass standoffs, channels, tie wire, indoor and outdoor enclosures, flex/flex connectors.

5) Emergency Lighting 1 2 3 4

Troubleshooting: BATTERY BACKUP BALLAST: Make sure the lamp is good/test the line voltage. If lamp is good press the test button if the lamp comes on when the button is pressed then the battery backup is good, replace the standard ballast. If the lamp(s) do not come on while pressing the test button then check the wiring/sockets and replace the battery backup if wiring is found to be good. BUG EYE/EXITS: press the test button if it does not come on test the line voltage, replace unit if voltage is good.

Components: battery backup ballasts/exits, bug eyes, emergency lighting circuit.

6) LED 1 2 3 4

Troubleshooting: If whole sign is out check line voltage. If voltage is good then test for secondary voltage off transformer. If secondary voltage is good test low voltage connections. If outages are sporadic test connections/swap modules that are out with known good modules to identify the bad ones. Always test and retest connections these are more often than not the issue. If secondary voltage out of the transformer is not good verify the secondary connections and replace the transformer. Some transformers have a cut out on them and will not read secondary power if they are not hooked up to any modules that is why it is very important to check and recheck the secondary connections. Components: LED modules, transformers, connectors, wire, straps/sticky tape, flex, outdoor/indoor enclosures.

6) Case Lighting 1 2 3 4

Troubleshooting: Check for switches, Mark location and come back in ½ hour if there is

potential for defrosting. Check the lamps, if not the lamps then check the sockets for breaks/burns. If the socket appears to be good then tear apart case and locate the ballast and test for proper line voltage. If line voltage is correct then verify the ballast is remote if it is not located in the hood or the door. Replace the ballast with the correct remote ballast for the lamp application. If ballast does not solve the problem then check the wiring harness continuity. If the wiring harness continuity can be repaired without breaking the seals make the repairs, if the seals must be broken to replace the harness then turn the service over to the manager to call a case specialist/refrigeration company. If there is not proper line voltage, verify the unit is not in defrosting mode/check for breakers/write it up for electrician.

Components: Case, lamps, sockets, ballasts, wire harness, reflectors, tube guards, lenses, distribution block.

7) Motion Sensors 1 2 3 4

Troubleshooting: If the switch works but is just not sensitive enough then check the setting on the back/side of switch. If the switch does not work at all first check for good ground as many motions sensing switches require this, then check the incoming power to the switch to verify good voltage across the hot and neutral. If the line voltage is good verify sensitivity settings and test switch leg to verify it is not sending power. If the settings are correct and the switch is still not sending power to the fixtures then replace the switch. For ceiling mounted sensors you need to isolate which component is bad either the power pack or the low voltage ceiling mounted sensor. First verify that the power pack is getting power and sending power to the sensor. If it is not sending power but is receiving it replace the power pack. If the power pack is sending low voltage power and the power is reaching the sensor replace the sensor.

Components: Fixture, power-pack, wire, low voltage wire, sensor

8) Time clock/Photo Cell 1 2 3 4

Troubleshooting: Verify clock is keeping time and spin the clock to verify the trippers are kicking on and off without the bypass replace trippers if needed. Listen for the clock motor, but don't rely solely on the sound to verify clock motor is working. Isolate line and load and verify that the line is hooked to the line side of the time clock or P/C. Verify that you have good line voltage across the hot and neutral. Cover P/C or turn time clock into bypass and verify voltage on the load side. If no load voltage, replace the time clock or P/C. Check for light sources near the photocell and adjust photocell sleeve depending on light source proximity/photocell orientation. Components: Trippers, clock, photocell, photocell sleeve, photocell receptacle

9) Data cable 1 2 3 4

Troubleshooting: Verify pin configuration on both sides of the cable (use cat5 pin chart). Verify the pins have been crimped properly. Use temporary patch cable (known good cable) to verify the computer/connected devices function with temp patch cable. Verify cat5 female receptacle is receiving the male end/making good connection with male end.

If devices work with temp cable replace with new permanent cable. 90% of the time the problem is with the crimp/pin configuration.
Components: Data cable, male and female terminals, computers/connecting cat5 devices.

10) Contactors 1 2 3 4

Troubleshooting: Verify line side voltage; verify coil voltage while controls are engaged. If you have line voltage and coil voltage (note: some coils engage when there is the absence of power) you should also have voltage on all load contact terminals. If no load voltage in this case write up to have an electrician replace the contactor. Check for good connections/burned and overheated wiring and note and wiring needs on the service order. Contactors can often times be rebuilt, but check with an electrician before writing up a rebuild or coil replacement.

Components: Contactor housing, contact points, coil, enclosure, wire

11) Breakers 1 2 3 4

Troubleshooting: Check for breaker amperage tolerance levels (do not open the dead front of a panel unless you have specific electrical training) and write it up for an electrician to replace breaker if it trips intermittently within the tolerance range. Check for loose connection/overheating of wire and breaker/panel. Wiggle switch arm. This can sometimes be an indicator of a weak breaker. Components: Panel, buss bar, breaker, wire

12) 3way switching 1 2 3 4

Troubleshooting: Wiggle switch arm, which can be an indicator of a bad switch. Check connection points on the switch for loose connections if the fixtures receive power intermittently. Use 3way switching diagram to trace power flow from switches to the fixtures or write it up for an electrician.

Components: Switches, wire, fixtures

13) Low voltage lighting systems 1 2 3 4

Troubleshooting: Check lamp continuity, check socket for burns/breakage/excessive wear/connection issues. Test line voltage across the hot and neutral. Test continuity between the socket and the transformer. If you have a good lamp a good socket and continuity as well as line voltage replace the transformer or fixture. If the problem is intermittent there could be connection issues inside the fixture. If no connection issues can be found and no parts were replaced to get the fixture working again-replace the fixture.

Components: Lamp, socket, fixture, transformer, low voltage cable/conductors, wire

14) Track Lighting 1 2 3 4

Troubleshooting: Verify there is at least one good lamp in the track. If outages are

sporadic then check for power at the sockets before replacing the lamps/ballasts. If there is no power at the socket but other fixtures on the track work then check for a switch on the track fixture/check for good connection to the track or move the fixture to a slightly different spot on the track. If moving the fixture does not correct the problem then test for continuity (incan or halogen only) between the track connection points on the fixture and the socket. If the fixture is ballasted then replace the ballast if other fixtures on the track are operating. If the entire track is out, check power/connections at the entrance fitting. Components: Track, track fixtures, lamps, ballasts, wire

15) Relays 1 2 3 4

Troubleshooting: Contact monitoring service for the facilities controls and troubleshoot over the phone with them until the problem is located. In most cases this requires an electrician. Components: wire, control board, relays, bypass

16) Outlets & Switches 1 2 3 4

Troubleshooting: Look for damaged, burned or loose parts. Check connections and replace if line voltage is good.
Components: Outlet, switch, wire, fixture, elec. box